

DCR2b: The Multi-Document Coverage Rule Fixes DCR2a's Placebo Failure

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Date-Cut Retrodiction – DCR2b (aggregation-rule repair) **Status:** Overall **G0**. All six N-gates pass. Multidoc scoring de-ranks T1 at the 1880 deep placebo (single Maxwell hit → score 0) while preserving T1 at the 1904 target cut (5-document coverage). Class-based nomination on the DCR arc now has a placebo-clean scoring rule. **Date:** 2026-07-27

Abstract

DCR2a found that cardinality-based class scoring put the T1 class at rank 2 at the 1880 deep placebo cut, because a single Maxwell 1865 hit inflated a low-cardinality class. Its licensed reading explicitly named the repair: *"aggregation rule must require multi-document coverage."* DCR2b applies exactly that repair – the multidoc rule de-ranks any class whose members come from fewer than a preregistered threshold of distinct documents (`min_docs = 2`).

Result: all six N-gates pass, overall G0. T1 at 1904 keeps its rank-3 position (with 5-document coverage under `target_v4`). T1 at 1880 scores 0 (correctly de-ranked). N3 G0: rank-3 beats proposition-blind rank-4 (same one-position gain DCR2a reported). N4 G0: T1 at 1880 is excluded from meaningful ranking. N6 G0: T1 at 1904 is not over-demoted.

Licensed reading: *"multidoc_fixes_placebo_failure: DCR2a's N4 defect is resolved. DCR2 pipeline can adopt multidoc as the aggregation rule."*

DCR2b's contribution is small – a single-rule repair on a corpus of existing results – but the loop it closes is load-bearing. DCR2a opened the specific question *"can any class-scoring rule survive both the target-cut and the placebo?"* and named the fix in advance. DCR2b implements exactly the named fix and reports the outcome.

1. What was preregistered

DCR2B_PREREGISTRATION.md (2026-07-27, before `nominate_by_multidoc.py` was drafted or `run_dcr2b.py` executed):

cardinality when the class covers at least 2 distinct documents; score = 0 otherwise.

baseline (word-count scoring). No edits to `nominate_by_class.py` or any prior module.

- **N4** (T1 silent at 1880) – must rank ≥ 3 **or** be de-ranked entirely (score 0). Preregistration explicitly allows either. - **N6** – multidoc must NOT demote T1 at 1904. Guards against a rule that is too restrictive.

- One aggregation rule: `multidoc(min_docs=2)`. Score = class
- Reuse of DCR2a's classification (`target_v4`) and proposition-blind
- The full N-suite from DCR2a, plus one new gate:
- Single-shot commitment: no replay knobs, no sweep over `min_docs`.

2. Results

Per-cut summary under `multidoc(min_docs=2)`:

cut	class	members	documents	multidoc score	rank
1880 (placebo)	unclassified	50	3	50	1
1880	T1_absolute_simultaneity	1	1	0	(de-ranked)

1880	T2_privileged_frame	0	0	0	(empty)
1880	T3_local_time_artifice	0	0	0	(empty)
1904 (target)	unclassified	235	15	235	1
1904	T2_privileged_frame	11	8	11	2
1904	T1_absolute_simultaneity	7	5	7	3
1904	T3_local_time_artifice	2	2	2	4

Proposition-blind baseline at 1904: best T1 realisation at rank 4 (same as DCR2a).

3. Gate decisions

gate

N1 class assignment complete	G0	235 / 254 unclassified at 1904 (< 95%)
N2 T1 class non-empty at 1904	G0	7 members $\geq$ 5
N3 multidoc beats proposition-blind	G0	rank 3 vs 4 (one-position gain, same as DCR2a)
N4 T1 de-ranked at 1880	G0	multidoc score 0 (singleton document)
N5 comparison table produced	G0	this section
N6 T1 at 1904 not over-demoted	G0	coverage 5 $\geq$ 2

Overall G0.

4. What the result says

The DCR2a defect was that cardinality-based scoring gave any nonempty class a strictly positive score, so a class with a single misfiring member could still outrank empty or near-empty classes at the placebo cut. The multidoc rule exchanges cardinality-of-members for cardinality-of-documents: score is cardinality only if the class covers at least `min_docs = 2` distinct documents, otherwise zero. Two consequences:

document (Maxwell 1865). Multidoc gives it a score of 0. It is correctly excluded from meaningful ranking.

distinct documents. Multidoc gives it a score of 7. It ranks 3, one position higher than the best proposition-blind T1 realisation ranks.

- **At the placebo cut (1880)**, the T1 class has one member from one
- **At the target cut (1904)**, the T1 class has 7 members from 5

The rank-2 position T1 receives at 1880 in the raw output is an alphabetical tiebreak among 0-score classes (T1 < T2 < T3), not a substantive ranking. Under any principled tiebreak or under the natural "exclude 0-score classes" reading, T1 is not ahead of anything at 1880.

5. What DCR2b establishes and does not establish

Establishes:

T1 class is placebo-clean under multidoc without losing its positive result at the target cut. the DCR series, a scoring rule that satisfies all six N-gates simultaneously.

- The DCR2a-preregistered repair works exactly as anticipated. The
- Class-based nomination on the DCR arc has, for the first time in

Does not establish:

corpus.** DR7 showed the wall relocates from N to g; multidoc is a rule on top of an already-imported g (= target_v4). DR7's soundness-completeness gap on open realisation sets is not addressed by any aggregation rule.

now be run.** DCR2b tested only the class-scoring layer, not the DR1-DR4 nominator pipeline. DCR2 remains open.

multi-document coverage.** Some legitimate targets in some corpora may have single-document realisations (e.g., a unique canonical statement of a mathematical convention appearing in exactly one paper). Multidoc would incorrectly de-rank them. For the DCR corpus, where T1 realisations are distributed across the electrodynamics literature by design, the rule fits.

- **Whether class-based nomination solves the DR5 wall on this
- **Whether DCR2 (the full deletion-repair nomination pipeline) can
- **Whether the rule generalises to targets without natural

6. Next work

directly, rather than as an intermediate step feeding target_v4. Would test whether a semantic-classifier grouping function catches T1 realisations target_v4 misses.

consensus with multidoc as the aggregation rule, and score against ground truth (Einstein's 1905 deletion). This is DCR2's original question, now with the aggregation rule fixed.

- **DCR2c** – the presupposition-inferring extractor used as \$g\$
- **DCR3** – run the DR-arc deletion-repair nominator over the DCR1e

Both are DCR-arc extensions. Neither is required to close DCR2b's loop; both are natural follow-ups DCR2b's G0 opens.

Appendix: reproduction

```
uv run --no-sync python -m experiments.date_cut_retrodiction.run_dcr2b
```

Reads the DCR1e presupposition-extraction consensus, applies
nominate_by_multidoc.score_multidoc at each of the three DCR cuts, compares to the
proposition-blind baseline, writes results/dcr2b_verdict.json. Local CPU, milliseconds.

Preregistration digest (SHA-256 of DCR2B_PREREGISTRATION.md):
547b5164361b648928b1aafcd09556e6468bc6224c9ba26eb22163304a886a78.